Sixty-nine years later, in 1998, science exhumed lambda one last time. Early that year, remarkable announcements were made by two competing teams of astrophysicists: one led by Saul Perlmutter of Lawrence Berkeley National Laboratory in Berkeley, California, and the other co-led by Brian Schmidt of Mount Stromlo and Siding Spring observatories in Canberra, Australia, and Adam Riess of the Johns Hopkins University in Baltimore, Maryland. Dozens of the most distant supernovas ever observed appeared noticeably dimmer than expected, given the well-documented behavior of this species of exploding star. Reconciliation required that either those distant supernovas behaved unlike their nearer brethren, or they were as much as fifteen percent farther away than where the prevailing cosmological models had placed them. The only known thing that "naturally" accounts for this acceleration is Einstein's lambda, the cosmological constant. When astrophysicists dusted it off and put it back into Einstein's original equations for general relativity, the known state of the universe matched the state of Einstein's equations.

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The supernovas used in Perlmutter's and Schmidt's studies are worth their weight in fusionable nuclei. Within certain limits, each of those stars explodes the same way, igniting the same amount of fuel, releasing the same titanic amount of energy in the same amount of time, thereby reaching the same peak luminosity. Thus they serve as a kind of yardstick, or "standard candle," for calculating cosmic distances to the galaxies in which they explode, out to the farthest reaches of the universe.

Standard candles simplify calculations immensely: since the supernovas all have the same wattage, the dim ones are far away and the bright ones are close by. After measuring their brightness (a simple task), you can tell exactly how far they are from you and from one another. If the luminosities of the supernovas were all different, you could not use brightness alone to tell how far away one was in comparison with another. A dim one could be either a high-wattage bulb far away or a low-wattage bulb close up.

All fine. But there's a second way to measure the distance to galaxies: their speed of recession from our Milky Way—recession that's part and parcel of the overall cosmic expansion. As Hubble was the first to show, the expanding universe makes distant objects race away from us